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Chapter 4

 Chapter 4 Proposed System Design (System Development Based) / Experimental Setup (Research Based)

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REQUIREMENT ANALYSIS AND DESIGN

Introduction

This chapter delineates the functional and non-functional requirements for the AI-Muneer Online Portal. It also elaborates on the system architecture, database schema, user interface design principles, and essential process flows. The main objective of this phase, carried out within PSM1, is to convert the stakeholder requirements and project goals identified in previous phases into a detailed blueprint for the system's development in PSM2. The design focuses on delivering an intuitive and effective experience for both potential clients (Visitors) seeking information about AI-Muneer Hall and the administrator (Owner) overseeing the portal's content and operations. The requirements and design components recorded in this document will serve as the foundation for the Software Requirements Specification (SRS) and System Design Document (SDD).

Requirement Analysis

Requirement analysis involves delineating the users' expectations and the system's characteristics. In the case of the AI-Muneer Online Portal, the requirements were primarily collected through direct engagement with the stakeholder (Mr. Almunajid, the proprietor of AI-Muneer Hall), as elaborated in the Case Study (Chapter 2.2.4).

Additionally, an examination of standard functionalities anticipated from comparable

online venue management systems was conducted, specifically adapted to the local context of Yemen.

Functional Requirements

Functional requirements describe what the system should do. For the Al-Muneer Online Portal, these are categorized by user roles and key system operations:

FR1: Visitor Functionalities

FR1.1 View Venue Information: The system shall display comprehensive information about Al-Muneer Hall, including descriptions, capacity, services offered, and location details.

FR1.2 View Media Gallery: The system shall allow visitors to browse a gallery of images and potentially videos showcasing the hall.

FR1.3 Check Availability: The system shall provide an interactive calendar allowing visitors to view dates that are already booked and those that are available.

FR1.4 View Pricing Information: The system shall display pricing information. This may include base rates and options for customizable packages or services that affect the price.

FR1.5 View FAQ: The system shall display a list of frequently asked questions and their answers.

FR1.6 Submit Booking Inquiry: The system shall provide a form for visitors to submit a booking inquiry, including their name, contact details, desired date(s), and any specific requests.

FR1.7 Submit Payment Proof (Optional): The system shall allow visitors, after submitting an inquiry or upon instruction from the admin, to optionally upload an image file (e.g., screenshot of a local transfer receipt) as proof of payment. The system shall only accept common image formats (e.g., JPG, PNG) and enforce a reasonable file size limit.

FR1.8 Submit Feedback: The system shall provide a form for visitors to submit feedback about the venue or their experience with the portal.

FR2: Administrator Functionalities

FR2.1 Secure Admin Login: The system shall require administrators to log in using unique, authenticated credentials (username and hashed password). The system shall display an error message if authentication fails.

FR2.2 Manage Venue Information: The system shall allow the administrator to create, update, and delete textual information about the hall (descriptions, services, FAQs).

FR2.3 Manage Media Gallery: The system shall allow the administrator to upload new images/videos to the gallery and delete existing ones.

FR2.4 Manage Availability Calendar: The system shall allow the administrator to mark dates as booked or available on the interactive calendar.

FR2.5 Manage Pricing: The system shall allow the administrator to set and update base pricing and any customizable pricing options.

FR2.6 Manage Booking Inquiries: The system shall allow the administrator to view submitted booking inquiries.

FR2.7 Manage Payment Proofs: The system shall allow the administrator to view uploaded payment proofs and update the status of a booking (e.g., "Payment Verified," "Pending Confirmation").

FR2.8 Manage Feedback: The system shall allow the administrator to view submitted user feedback.

FR2.9 Generate Basic Reports: The system shall allow the administrator to view basic reports, such as a list of inquiries within a date range, a summary of booking statuses, or an overview of feedback received.

FR2.10 Manage Notifications: The system shall allow the administrator to configure triggers or templates for notifications (e.g., inquiry received, payment proof acknowledged) to be sent via preferred channels (SMS/WhatsApp, with email as a secondary option).

FR3: General System Functionalities

FR3.1 Responsive Design: The system shall be responsive and accessible on common web browsers across various devices (desktops, tablets, smartphones).

FR3.2 Secure Data Handling: The system shall securely handle all user-submitted data, including personal information in inquiries, uploaded payment proofs, and feedback.

FR3.3 Notification Generation: Based on admin configuration and specific system events (e.g., new inquiry, payment proof submission), the system shall generate notifications to be sent to relevant parties (client and/or admin).

Non-Functional Requirements

Non-functional requirements define the quality attributes of the system.

NFR1 Performance: The system should load pages within an acceptable time frame (e.g., 3-5 seconds on a typical internet connection). The interactive calendar should respond promptly to user interactions.

NFR2 Usability: The portal should be intuitive and easy to navigate for both visitors and the administrator. The language used will be primarily Arabic, with an option for English if deemed feasible within the project scope. UI elements should be clear and user-friendly, considering local cultural norms for design.

NFR3 Security:

NFR3.1 Admin Authentication: The admin login shall be protected against common vulnerabilities. Passwords must be hashed.

NFR3.2 Data Protection: User-submitted data and uploaded files (payment proofs) must be protected from unauthorized access.

NFR3.3 SSL/TLS: All communication between the client and server shall be encrypted using HTTPS.

NFR4 Reliability: The system should be available for use with minimal downtime.

Database backups should be considered as part of operational reliability.

NFR5 Maintainability: The code should be well-structured, commented, and follow the chosen MVC pattern to facilitate future updates or modifications (relevant for PSM2 and beyond).

NFR6 Scalability (Basic): While not expecting massive concurrent users initially, the system architecture (Spring Boot, chosen database) should be capable of handling a moderate increase in traffic (up to 10,000 monthly visitors as per initial scope) without significant performance degradation.

Use Case Table and Diagram

Selected Use Case: Submit Booking Inquiry (UC005)

Attribute

Description

Use Case ID

UC005

Use Case Name

Submit Booking Inquiry

Actor(s)

Visitor

Pre-condition(s)

Visitor is on the Al-Muneer Online Portal and has navigated to the inquiry section/form.

Normal Flow

Visitor accesses the booking inquiry form.

Visitor fills in required fields (e.g., name, contact, desired date, event type, number of guests, specific requests).

Visitor submits the form.

System validates the input data.

System saves the inquiry details to the database.

System displays a success message to the Visitor.

System triggers a notification to the Administrator about the new inquiry.

Alternative Flow

4a. Invalid Input Data:

4a1. System displays an error message indicating the fields with invalid data.

4a2. Visitor corrects the data and re-submits (back to step 3).

7a. Notification Failure:

7a1. System logs the notification failure but still saves the inquiry and shows success to the visitor.

Admin will see the inquiry upon login.

Post-condition(s)

Booking inquiry is successfully recorded in the system.

Administrator is notified of the new inquiry.

Visitor receives confirmation that their inquiry has been submitted.

Table 4.1: Use Case Description for 'Submit Booking Inquiry' (UC005)

Sequence Diagram (Submit Booking Inquiry - UC005)

Activity Diagram (Submit Booking Inquiry - UC005)

Project Design

The Al-Muneer Online Portal utilizes a Model-View-Controller (MVC) architectural framework within a singular, monolithic application. This structure guarantees a clear delineation of responsibilities: the Model (which encompasses data and business logic), the View (representing the user interface), and the Controller (which manages user input and the flow of the application). This cohesive strategy is exceptionally well-suited to the project's needs, facilitating both development and deployment.

System Architecture Diagram

The diagram presented below depicts the overarching Model-View-Controller (MVC) architecture utilized by the Al-Muneer Online Portal. Within this unified framework, the

Spring Boot application is responsible for managing both the business logic and the rendering of the user interface, thereby establishing a coherent and sustainable system.

Figure 4.4 System Architecture Diagram (MVC) //todo update me in table

Database Design

A relational database model is selected because of the organized characteristics of the data. The subsequent illustration is a basic conceptual Entity-Relationship Diagram (ERD). Primary Keys (PK) and Foreign Keys (FK) will be specified in a comprehensive design.

Data Components:

AdminUser: Stores admin credentials and roles.

VenueInfo: General descriptive information about the hall.

MediaItem: Details about uploaded images and videos for the gallery.

AvailabilitySlot: Represents specific dates and their booking status.

PricingTier: Information about different pricing packages or rates.

BookingInquiry: Details of inquiries submitted by visitors.

PaymentProof: Metadata for uploaded payment proof files, linked to inquiries.

Feedback: Stores feedback submitted by users.

ReportConfig: (Conceptual) Could store metadata about generated reports if needed for history, though simple reports might be generated on-the-fly.

Interface Design (Low-Fidelity)

Homepage:

Purpose: Welcome visitors, provide a brief overview of the hall, high-quality hero image/video, clear navigation to other sections (Gallery, Availability, Pricing, Inquiry, FAQ, Feedback).

Key Elements: Navigation bar, hero section, brief intro text, call-to-action buttons.

Gallery Page:

Purpose: Showcase the venue through images and videos.

Key Elements: Grid or carousel of media items, ability to view larger images/play videos.

Availability Page:

Purpose: Allow visitors to check for available dates.

Key Elements: Interactive calendar highlighting booked/available dates, potentially a date picker to jump to specific months/years.

Pricing Page:

Purpose: Provide clear information about pricing structures and packages.

Key Elements: Sections for different packages/services, details of what's included, base prices.

Note: These are dummy pricings and not accurate. The stakeholder will have an access to handle and set all the pricings.

Inquiry Page:

Purpose: Allow visitors to submit detailed booking inquiries.

Key Elements: Form with fields for name, contact, event date, number of guests, message. Clear submit button.

Payment Proof Upload Page (May be part of inquiry follow-up or a separate link):

Purpose: Allow users to upload their payment proof securely.

Key Elements: File upload input, instructions on accepted formats/sizes, submit button.

Feedback Page:

Purpose: Collect feedback from visitors.

Key Elements: Form for name (optional), contact (optional), feedback text, rating (optional), submit button.

Chapter Summary

This chapter outlined the fundamental results of the requirement analysis and system design stages for the Al-Muneer Online Portal. It began by specifying the identified functional requirements, which delineate the portal's functionalities for both users (e.g., viewing venue information, checking availability, submitting inquiries and payment confirmations, providing feedback) and the administrator (e.g., content management, processing inquiries, generating reports). Essential non-functional requirements concerning performance, usability, security, reliability, maintainability, and basic scalability were also articulated. The chapter subsequently described the system's layered architectural framework, demonstrating the interaction among the presentation, application, and data layers. A conceptual Entity-Relationship Diagram was included for the relational database, outlining the primary data entities. User interactions were further elucidated through a detailed use case diagram, with the "Submit Booking Inquiry" use case being elaborated upon via a textual description, sequence diagram, and activity diagram to illustrate system flow and behavior. Conceptual designs for the user interfaces of key portal pages were also discussed, emphasizing user-friendliness and functional clarity. The work presented serves as the foundation for the SRS and SDD documents.